

Advanced (Carolina Reaper)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.

- (A) $x = -2, x = -3$
 (B) $x = -1, x = -6$
 (C) $x = 2, x = 3$
 (D) $x = 1, x = 6$
 (E) $x = -2, x = 3$

Q2. If $x^2 - 7x - 18 = 0$, what are the solutions?

- (A) $x = -1, x = 18$
 (B) $x = 9, x = -2$
 (C) $x = -9, x = 2$
 (D) $x = 1, x = 18$
 (E) $x = -2, x = 9$

Q3. Tom has been saving money for a new bike and has 120 in his savings account. He wants to buy a bike that costs 180. If Tom can save an additional x dollars per week, how many weeks will it take him to have enough money to buy the bike, given by the equation $x^2 + 2x - 60 = 0$?

- (A) Tom will take 5 weeks to save enough money.
 (B) Tom will take 6 weeks to save enough money.
 (C) Tom will take 8 weeks to save enough money.
 (D) Tom will take 10 weeks to save enough money.
 (E) Tom will take 12 weeks to save enough money.

Q4. A bookshelf has 5 shelves, and each shelf can hold 8 boxes. If the bookshelf is currently empty, how many boxes can be placed on it in total, modeled by the equation $5x + 8y = 40$ where x is the number of boxes on each shelf and y is the number of shelves?

- (A) 40 boxes
 (B) 50 boxes
 (C) 60 boxes

(D) 40 boxes

(E) 80 boxes

Q5. Factor the expression $x^2 - 4x - 21$.

- (A) $(x - 7)(x + 3)$
 (B) $(x + 7)(x - 3)$
 (C) $(x - 3)(x + 7)$
 (D) $(x + 3)(x - 7)$
 (E) $(x - 2)(x - 9)$

Q6. Solve for x in the equation $(x - 2)(x + 5) = 0$.

- (A) $x = 2$
 (B) $x = -5$
 (C) $x = 2, x = -5$
 (D) $x = 7$
 (E) $x = 3$

Q7. A rectangular garden measures 10 meters by 8 meters. If a path that is x meters wide is built around the garden, the area of the path is given by the equation $4x^2 + 36x + 80 = 0$. What is the width of the path?

- (A) $x = 2$ meters
 (B) $x = 4$ meters
 (C) $x = 5$ meters
 (D) $x = 8$ meters
 (E) $x = 10$ meters

Q8. If $x^2 - 9x + 20 = 0$, what are the solutions?

- (A) $x = 4, x = 5$
 (B) $x = 5, x = 4$
 (C) $x = 4, x = 20$
 (D) $x = -4, x = -5$
 (E) $x = 9, x = 20$

Open Ended Questions (FRQ)

Q9. In a small town, there is a skate park with a rectangular skate floor with a length of x meters and a width of 10 meters. If the area of the skate floor is 150 square meters, and x can be found by solving the equation $x^2 - 10x - 150 = 0$, find the dimensions of the skate floor and calculate the perimeter. Show your work and provide the final answer in simplest terms.

Q10. Solve the quadratic equation $2x^2 + 5x - 3 = 0$ by factoring. Then verify the solutions by plugging them back into the original equation. Provide the solutions and the verification steps.

Chef's Tips

- Make sure students show their work for all problems.
- Emphasize the importance of verifying solutions.
- Remind students to check their units and ensure they are consistent throughout their calculations.